

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIPE504A - ADVANCED DATABASE SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 A relation is in 2NF if it is in 1NF and has no:
a) Transitive dependency b) Multivalued dependency
c) Partial dependency d) Join dependency
 - 2 A weak entity set must have a _____ relationship with its owner entity.
a) Identifying b) Ternary c) Recursive d) Aggregated
 - 3 A relation is in BCNF if for every non-trivial functional dependency $X \rightarrow Y$, X is a _____.
a) Foreign key b) Super key c) Composite key d) Partial key
 - 4 The process of converting an ER diagram into relational tables is known as:
a) Normalization b) Mapping c) Aggregation d) Generalization
 - 5 Which join returns all records from the left table and matched records from the right table?
a) Inner Join b) Full Join c) Left Outer Join d) Cross Join
 - 6 In ER modeling, a super key that has no proper subset as a super key is called:
a) Foreign key b) Alternate key c) Candidate key d) Composite key
 - 7 In association rule mining, confidence measures the _____ of the rule.
a) Frequency b) Support c) Accuracy d) Conditional probability
 - 8 A schedule is conflict-serializable if its precedence graph is:
a) Cyclic b) Weighted c) Directed d) Acyclic
 - 9 The SQL command used to remove all rows from a table but retain its structure is:
a) DELETE b) DROP c) REMOVE d) TRUNCATE
 - 10 In timestamp ordering, a transaction is rolled back if it violates:
a) Lock compatibility b) Write-read rule
c) Timestamp ordering rule d) Commit rule
 - 11 The Apriori algorithm reduces the search space by exploiting:
a) Partitioning property b) Downward closure property
c) Hash indexing d) Clustering principle

- 12 In lossless-join decomposition of R into R1 and R2, the common attribute must be:
- Multivalued
 - Non-prime
 - A super key of at least one relation
 - A foreign key
- Q2. Solve the following.
- Draw an Entity–Relationship (ER) diagram for a University Management System that includes the entities Student, Course, Faculty, and Enrollment, and clearly indicate the primary keys, foreign keys, and the cardinality of relationships between them. 6
 - State and explain Codd's 12 rules for relational database systems. 6
- Q3. Solve the following.
- Define transaction control statements in SQL and explain COMMIT, ROLLBACK, and SAVEPOINT. 6
 - CUSTOMER(CustID INT PRIMARY KEY, Name VARCHAR(50), City VARCHAR(30))
ORDERS(OrderID INT PRIMARY KEY, CustID INT, Amount DECIMAL(10,2), FOREIGN KEY (CustID) REFERENCES CUSTOMER(CustID))
Write SQL queries to:
a) Retrieve customers who have placed orders above ₹10,000.
b) Compute total order amount per customer. 6
- Q4. Solve Any Two of the following.
- Explain the steps involved in the database design process from requirements analysis to normalization. 6
 - Explain Multivalued Dependency (MVD) and Fourth Normal Form (4NF) with formal notation. 6
 - Write a JDBC program in Java to:
Establish a connection to a MySQL database, insert a record into EMPLOYEE(EmpID, Name, Salary), and commit the transaction. Explain each step briefly. 6
- Q5. Solve Any Two of the following.
- Explain the architecture of a Data Warehouse system, including data sources, ETL process, staging area, and presentation layer. 6
 - Apply the K-means clustering algorithm ($k = 2$) on the dataset: (2, 3), (3, 4), (4, 5), (1, 2), (10, 12), (11, 13), (12, 14), (9, 11). Perform one iteration and show cluster assignment. 6
 - Explain the role of indexing and ranking mechanisms in information retrieval systems. 6
- Q6. Solve Any Two of the following.
- Explain the difference between recoverable, cascadeless, and strict schedules using appropriate examples. 6
 - Define conflict serializability and view serializability with formal notation. Consider the following schedule:
S1:R1(X), W1(X), R2(X), W2(X), R1(Y), W1(Y), R2(Y), W2(Y)
(a) Construct the precedence graph.
(b) Determine whether the schedule is conflict-serializable. 6

- C) Define deadlock and explain the necessary conditions for deadlock occurrence (Coffman conditions). Construct the wait-for graph for the following lock table scenario: 6

T1 holds X-lock on A; T2 requests X-lock on A.

T2 holds X-lock on B; T3 requests X-lock on B.

T3 holds X-lock on C; T1 requests X-lock on C.

Determine whether deadlock exists.

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